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SURVEY

Face Swap Detection: A Systematic Literature Review

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ABSTRACT Face swap technology, often associated with deepfakes, has rapidly advanced in recent years, raising serious concerns around misinformation, digital impersonation, and privacy. As a result, the development of reliable face swap detection methods has become a critical area of research. This survey provides a comprehensive review of existing approaches to face swap detection, addressing key research questions such as commonly used datasets, evaluation metrics, comparative model performance, and persistent challenges in the field. It includes a detailed taxonomy of detection methods, categorizing approaches into spatial, temporal, and spatiotemporal techniques. The survey further examines cross-dataset generalization performance to assess how well models adapt to domain shifts between training and testing data. Recent innovative directions are explored, covering adversarial defense strategies, explainability techniques, lightweight models for edge deployment, and privacy-preserving training. Additionally, best practices for building and releasing face swap detection tools are discussed to promote ethical, robust, and practical implementations. Finally, the paper outlines future research directions aimed at advancing model robustness, generalization, and compliance with legal and ethical standards. The discussions provide valuable insights that help researchers and practitioners gain a clear understanding of the face swap detection field, supporting and guiding their future research efforts.

INDEX TERMS Face swap detection, deepfake detection, adversarial attacks, deepfake evaluation metrics, digital forensics.

I. INTRODUCTION

In recent years, the advancement of artificial intelligence and deep learning technologies has led to the rapid emergence of deepfakes, synthetic media in which a person's likeness, voice, or actions are manipulated to appear realistic but are entirely or partially fabricated. Deepfakes have gained widespread attention due to their potential to mislead, deceive, and compromise trust in visual media. While some applications of deepfakes are harmless or even creative, such as in entertainment or education, their misuse has raised serious concerns related to misinformation, identity fraud, political manipulation, and personal privacy violations [1], [2].

Deepfake technologies include a variety of manipulation techniques, each posing unique challenges and implications. Lip-synching is one such method, where the mouth movements

of a person are altered to match a different audio track, creating the illusion that they are saying something they never actually said. This technique is particularly concerning when used to fabricate statements by public figures, potentially spreading misinformation or damaging reputations [3]. Another widely used approach is facial reenactment, which transfers the facial expressions and movements of one person onto another's face, often in real time. This manipulation can mimic emotions and gestures with high accuracy and is sometimes used maliciously in live or recorded media [4]. In contrast, entire face synthesis involves generating completely artificial, photorealistic faces that do not correspond to any real individual. These are typically created using Generative Adversarial Networks (GANs) like StyleGAN and are often used in fake online identities or misleading visual content [5].

Lastly, face swapping is one of the most recognizable forms of deepfake manipulation, where the face of one person is replaced with another's while maintaining realistic

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facial expressions, head movements, and lighting conditions. This type of deepfake is particularly deceptive due to its visual believability and is often used in harmful scenarios such as impersonation, fake pornography, or falsified video evidence [1]. As deepfake methods continue to evolve, it becomes increasingly critical to develop targeted detection techniques that address each specific manipulation type.

As the quality of face swap technology continues to improve, the need for specialized detection methods becomes more critical. However, much of the existing research and survey literature focuses broadly on deepfake detection without giving sufficient attention to the face swap detection methods. This study aims to address that gap by providing a focused investigation into face swap detection methods and the current state of research in this specific domain between the year 2020 and 2024. The following are the key contributions of this paper:

- **Comprehensive Dataset Review:** The survey provides a detailed overview of the most widely used datasets in face swap detection research. The analysis includes dataset characteristics, data distribution, and relevance to real-world detection scenarios.
- **Taxonomy of Face Swap Detection Methods:** The paper categorizes and analyzes the most common methodologies in the field of face swap detection, identifying dominant trends and highlight underexplored approaches, grouped into three categories: spatial, temporal, and spatiotemporal.
- **Evaluation Metric Analysis:** It systematically examines the performance metrics commonly used in the literature such as accuracy, precision, recall, area under the curve, and F1-score highlighting their strengths, limitations, and implications for comparing model performance.
- **Model Performance Comparison:** It presents a comparative evaluation of state-of-the-art face swap detection models, offering insights into their effectiveness across different datasets.
- **Cross-Dataset Generalization Performance:** The paper evaluates how well existing models generalize across different datasets and investigates the effects of domain shift. It exposes the limitations in current models' adaptability and highlight the need for improved robustness.
- **Discussion of Real-World Challenges:** A significant portion of the paper is dedicated to discussing the current challenges and limitations in face swap detection, including issues such as generalization gaps, adversarial attacks, data imbalance, and ethical and legal concerns.
- **Innovative Directions:** This survey explores recent innovations in face swap detection models that enhance robustness, explainability, and deployment readiness, such as adversarial defenses, model transparency, lightweight architectures for edge devices, and privacy-preserving training.

- **Best Practices for Tool Development and Release:** It proposes actionable guidelines for developing and deploying face swap detection tools responsibly. The aim is to ensure accuracy, fairness, security, and legal compliance in practical applications.
- **Future Research Directions:** The paper outlines strategic areas for future investigation to advance the field. This guides researchers toward emerging challenges and promising technologies, ensuring continued progress in building robust and ethical detection systems.

Beyond the scope of the core contributions of this paper, it is also important to highlight emerging directions that complement traditional face swap detection methods. One such direction involves the integration of person re-identification (Re-ID) techniques. Person Re-ID focuses on verifying whether the same individual appears across different videos, frames, or camera views, and offers a promising avenue for enhancing identity verification in deepfake detection. Incorporating Re-ID features can help systems detect inconsistencies between the claimed identity and the visual content, thus improving the robustness of face swap detection against identity manipulation. Several studies have proposed leveraging Re-ID embeddings or identity consistency checks as part of fake detection pipelines [6], [7]. Highlighting this connection opens new possibilities for future research at the intersection of biometric recognition and face swap detection, particularly in applications such as forensics, surveillance, and identity protection.

The structure of this paper is organized as follows: Section II provides an overview of existing surveys in the broader deepfake domain. Section III presents the methodology of this study, outlines the criteria used to collect and analyze the relevant literature, including the formulation of the key research questions that guide the survey. The Results and Discussion presents in Section IV. Finally, the Conclusion in Section V summarizes the key findings and contributions of the paper.

II. RELATED WORK

This section presents surveys and systematic literature reviews (SLRs) in the field of face swap detection. However, since no surveys or SLRs specifically focus on face swap detection, existing surveys in the broader domain of deepfake detection are summarized. The selected surveys are journal papers that are published in reputable sources including ACM, IEEE, Springer, and ScienceDirect.

The authors of [3] provide a comprehensive survey on deepfake technology, covering both deepfake creation algorithms and detection methods. The paper discusses the increasing sophistication of deepfake generation using deep learning models, particularly autoencoders and GANs. The paper also highlights the growing threats posed by deepfakes, including privacy violations, misinformation, and security risks.

On the detection side, the survey categorizes various deepfake detection techniques into image-based and

video-based approaches. It examines handcrafted feature-based methods, deep learning-based detection, and temporal feature analysis for video deepfakes. The study also addresses current challenges, limitations, and future research directions, emphasizing the need for more robust, generalizable, and explainable AI-based detection methods to counter increasingly sophisticated deepfakes.

The authors of [8] provide a comprehensive review of deepfake detection methods, emphasizing their reliability in real-world applications such as forensic investigations and legal cases. The paper explores the evolution of deepfake generation, from early autoencoder-based methods to advanced GAN-based techniques like StyleGAN and FaceShifter. The study also highlights key challenges in deepfake detection, including transferability (models struggle with unseen deepfakes), interpretability (lack of explainability in AI models), and robustness (vulnerability to compression, noise, and adversarial attacks). Evaluations of state-of-the-art detection models, such as Xception and MAT, reveal that while they perform well on specific datasets, their accuracy drops significantly when applied to real-world deepfakes.

To address these issues, the paper introduces a statistical reliability metric to assess detection model consistency and advocates for cross-dataset generalization to improve deepfake detection on previously unseen manipulations. It also calls for explainable AI methods to provide justifications for deepfake classifications and highlights the importance of legal validation for forensic use. The study concludes that existing deepfake detection models are not yet fully reliable, and future research should focus on developing robust, interpretable, and scalable solutions to counter increasingly sophisticated deepfake attacks.

In [9], they provides a comprehensive analysis of deepfake video generation and detection, emphasizing the growing societal concerns posed by AI-generated fake media. The paper categorizes the challenges in deepfake detection into data-related issues (imbalanced datasets, lack of labeled data), training difficulties (high computational requirements), and reliability concerns (overconfidence in detection models and evolving deepfake manipulation techniques). The study highlights the dominance of deep learning-based detection methods, which, despite their accuracy, suffer from limited generalization and high computational costs. The research also underscores the importance of high-quality datasets to improve detection capabilities and presents major gaps in existing approaches, advocating for the development of more robust, real-time detection models.

Furthermore, the paper compares deepfake image vs. video detection, noting that video deepfake detection is more complex due to temporal consistency and audio-visual synchronization challenges. It evaluates different detection techniques, including Convolutional Neural Network (CNN)-based models, Recurrent Neural Networks (RNNs), and hybrid approaches, and discusses the need for real-time, efficient, and explainable AI-based methods.

In [10], the authors present an extensive analysis of various methods for detecting deepfake videos. The paper categorizes detection techniques into deep learning-based approaches, classical machine learning methods, statistical techniques, and blockchain-based strategies. The study highlights that deep learning models, particularly CNNs, dominate deepfake detection due to their superior feature extraction capabilities. Additionally, the research identifies key datasets used for evaluation, such as the FaceForensics++ dataset, and emphasizes the importance of detection accuracy as the primary performance metric.

In [4], they provide a comprehensive survey on deepfake detection for human face images and videos. The paper discusses the fundamental concepts of deepfakes, including their applications and threats, particularly focusing on deep learning-based GANs used to create synthetic content. The study extensively examines various deepfake detection techniques, categorizing them into traditional forensic methods, deep learning-based approaches, and artifact analysis techniques. Traditional methods involve detecting inconsistencies at the pixel level, while deep learning models rely on spatial and temporal artifacts introduced during the generation process. The paper also highlights biometric-based detection techniques, such as analyzing facial expressions and physiological signals. Additionally, the study identifies key challenges in deepfake detection, including the lack of generalizability across different datasets, and the vulnerability of detection models to adversarial attacks.

On the other hand, the authors of [11] provide an in-depth exploration of deepfake technology, detailing both its creation and detection methods. They discuss various types of deepfakes, including reenactment and face replacement, which pose significant ethical and security concerns. While deepfake technology has legitimate applications, it has also been exploited for misinformation, political manipulation, and identity fraud. The paper reviews how GANs and autoencoders contribute to deepfake synthesis, highlighting their rapid evolution and growing accessibility.

On the detection front, the paper examines current techniques designed to identify deepfakes, including supervised learning models, anomaly detection methods, and adversarial defenses. However, it highlights the limitations of these methods, such as susceptibility to adversarial attacks and lack of generalizability across datasets. The study further discusses the challenges in deepfake detection, including the need for high-quality datasets, the computational costs of detection models, and the continuous improvement of deepfake generation techniques that outpace detection efforts. The survey concludes by outlining future research directions, emphasizing the need for robust detection frameworks, real-time deepfake mitigation strategies, and the exploration of the weaknesses and limitation of the current detectors.

The authors of [12] provide a comprehensive overview of deepfake technologies, specifically focusing on face swapping and expression re-enactment techniques. The

paper categorizes face manipulations into face generation, facial attribute changes, face swapping, and expression swapping. It details various generation techniques, including subject-specific and subject-agnostic approaches like FaceShifter, SimSwap, and FSGAN, and discusses their limitations in terms of computational cost, identity leakage, and deepfake quality. The authors also review a wide range of deepfake datasets across three generations, with newer datasets providing higher quality, diversity, and real-world perturbations. In terms of detection, the paper examines artifact-based, pixel and statistical based, general neural network, anomaly based, spatiotemporal, localization approaches. Finally, it outlines key challenges and future research directions, including improving generation quality, preserving identity under occlusions, developing interpretable and robust detectors, and expanding datasets for real-world applications.

In conclusion, the reviewed surveys offer valuable overviews of deepfake detection techniques, addressing a wide range of manipulated media formats. However, most of these studies take a general approach and do not specifically focus on face swap detection. In contrast, the current work provides a targeted analysis dedicated exclusively to face swap manipulations. Table 1 presents a detailed comparison between this study and existing surveys, highlighting key differences in scope, and contributions. This comparison reveals a significant gap in the literature, a lack of comprehensive and focused analysis of face swap detection methods, which this work aims to address.

III. METHODOLOGY

The section outlines the systematic approach followed in conducting this study. This section is structured into several key subsections, each addressing a critical aspect of the methodology. It begins with the formulation of the search string and identification of relevant digital libraries used to retrieve literature. Next, the inclusion and exclusion criteria are defined to filter studies based on their relevance and quality. A quality assessment framework is then applied to evaluate the reliability of the selected studies. The PRISMA diagram is presented to visually summarize the selection process. Finally, the research questions that guide this study are detailed, providing a clear focus for the analysis that follows.

A. SEARCH STRING AND LIBRARIES

Four different libraries are used to collect the papers including IEEEExplore, ACM Digital Library, SpringerLink, and Science Direct. The Search String that are used to collect the data is ((Face Swap OR Fake Face) AND Detection). This search string is used to be searched in the title and the abstract of the papers.

B. INCLUSION AND EXCLUSION CRITERIA

The papers are filtered based on a set of inclusion and exclusion criteria as follows:

- Inclusion criteria:
 - 1) Research studies published from 2020 to 2024.
 - 2) Conference or journal paper.
 - 3) The study must be a full-length paper.
- Exclusion criteria:
 - 1) Papers written in languages other than English have been excluded.
 - 2) The review or survey papers are excluded.
 - 3) Paper that publish a dataset.

C. QUALITY ASSESSMENT

The papers identified based on the inclusion and exclusion criteria will undergo a Quality Assessment process. The following are the Quality Assessment criteria:

- **QA1:** The paper must specifically address the face swap detection problem rather than general deepfake or face forgery detection.
- **QA2:** The paper should focus solely on visual data for forgery detection, excluding methods that rely on audio or something else.
- **QA3:** The paper must include an evaluation of the proposed approach, with quantitative metrics.
- **QA4:** The dataset used for evaluation should be publicly available or described in sufficient detail for reproducibility.

D. PRISMA DIAGRAM

The PRISMA diagram, depicted in Figure 1, provides a comprehensive summary of the systematic process followed in this survey. During the identification phase, a total of 456 records were retrieved through database searches. In the screening phase, the records were reviewed based on their titles and abstracts, using the exclusion criteria outlined in Section III-B. This step ensured the elimination of irrelevant studies, resulting in a total of 405 papers proceeding to the next stage. Subsequently, in the eligibility phase, the records were subjected to full-text assessment, guided by the quality evaluation process described earlier in Section III-C. Ultimately, 161 papers met the inclusion criteria and were incorporated into this research. These papers are distributed across key publishing platforms as follows: IEEE (74 papers), Springer (39 papers), Science Direct (27 papers), and ACM (21 papers). The distribution of the papers based on the years and publisher presents in Figure 2.

E. RESEARCH QUESTIONS

This research aims to answer the following questions:

- RQ1** : Which datasets are commonly used for training and evaluating face swap detection methods?
- A1** : This question aims to catalog the datasets available in face swap detection research, as well as their relevance and role in model training and evaluation. Examples of Face Swap detection datasets are Celeb-DF, FaceForensics++, and DeepFake Detection Challenge datasets.

TABLE 1. Comparative analysis of related survey papers.

Paper	Deepfake Scope	Generation Techniques Reviewed	Detection Techniques Reviewed	Datasets Discussed	Performance Metrics Discussed	Challenges Highlighted	Main Contributions
Nguyen et al. (2022) [3]	Deepfake in general	Thoroughly discussed	Thoroughly discussed	Not discussed	Not discussed	General discussion	Broad overview of deepfake landscape
Wang et al. (2024) [8]	Deepfake in general	Briefly mentioned	Present main papers in the field from reliability perspective	Briefly discussed	Not discussed	Focus on transferability, interpretability, and robustness	Focus on reliability-oriented deepfake detection approaches
Kaur et al. (2024) [9]	Deepfake in general	Thoroughly discussed	Present the papers in the field	Briefly discussed	Discussed	Present data-driven challenges	Classification of data-driven challenges in deepfake detection approaches
Rana et al. (2022) [10]	Deepfake in general	Briefly mentioned	Thoroughly analyzed papers published between 2018 - 2020	Briefly discussed	Discussed	Not discussed	Extensive classification of detection methods
Malik et al. (2022) [4]	Deepfake in general	Thoroughly discussed	Thoroughly discussed	Discussed forgeries detection datasets	Not discussed	Focus on lack of deepfake datasets, unknown type of attack, temporal aggregation, and unlabeled data	Deepfake detection and generation methods discussion
Mirsky et al. (2021) [11]	Deepfake in general	Thoroughly discussed	Thoroughly discussed	Not discussed	Not discussed	Briefly discussed	Focus on how deep-fakes are created and detected
Waseem et al. (2023) [12]	Face swap and expression re-enactment	Thoroughly discussed	Thoroughly discussed	Thoroughly discussed	Not discussed	General discussion	Deep dive into face swap and expression swap methods
Current Work	Face swap detection	Briefly mentioned	Thoroughly analyzed papers published between 2020 - 2024	Thoroughly discussed	Thoroughly discussed	Focus on variability in image and video quality, data imbalance, generalization to unseen data, adversarial manipulation, and privacy and legal issues	Tailored exclusively on face swap detection with analysis of models, datasets, metrics, and challenges

RQ2 : What techniques and methodologies are most commonly employed in face swap detection research?

A2 : The aim of this research question is to identify and categorize the most commonly used techniques and methodologies within the field of face swap detection. Understanding which approaches dominate the literature helps to map the current state of research, reveal trends, and highlight potential gaps for future investigation. Generally, studies in face swap detection can be categorized into three main types: spatial, temporal, and spatiotemporal approaches.

RQ3 : What performance metrics and evaluation measures are typically applied in face swap detection research?

A3 : This question aims to identify the key metrics and measures used to evaluate the success and accuracy of face swap detection models. It seeks to understand how different models are assessed and compared. Commonly used metrics are such as accuracy, precision, recall, F1-score, and area under the curve.

RQ4 : How do different face swap detection models compare in terms of accuracy, precision, recall, and other relevant performance indicators?

A4 : This question seeks to compare the effectiveness of various face swap detection models by examining their performance on different evaluation metrics.

RQ5 : To what extent do state-of-the-art face swap detection models generalize across different datasets, and how does domain shift between training and testing data impact their performance?

A5 : The aim of this research question is to investigate the cross-dataset generalization capabilities of leading face swap detection models and to analyze the effect of domain shift on their performance. This will help identify the limitations of current approaches and highlight the need for more robust, adaptable detection methods.

RQ6 : What are the challenges and limitations faced in face swap detection?

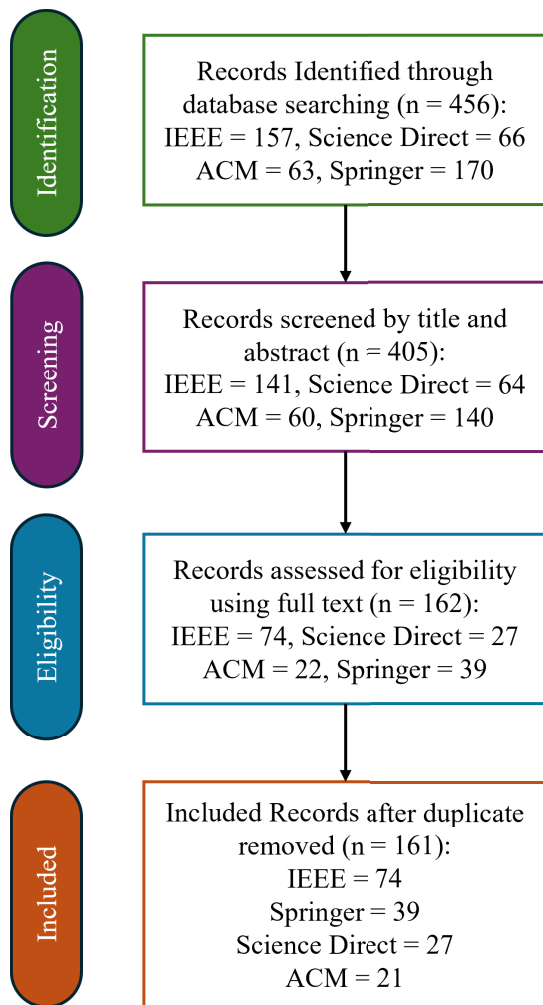


FIGURE 1. PRISMA diagram.

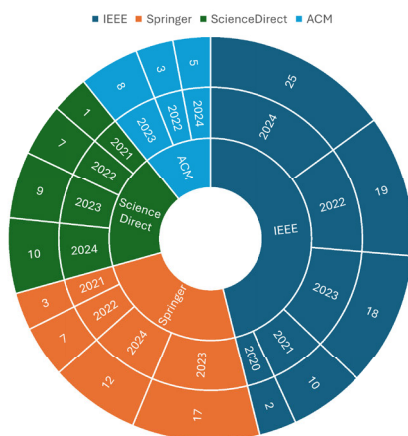


FIGURE 2. Distribution of papers based on the publisher and the year.

A7 : The aim of this question is to review and evaluate emerging methods that address key limitations in face swap detection, with a focus on practicality, security, and ethical deployment such as adversarial defense, explainability, edge optimization, and privacy-preserving training.

RQ8 : what best practices should be followed when building and releasing face swap detection tools?

A8 : The aim of this question is to identify and establish comprehensive guidelines and strategies that ensure face swap detection tools are developed with high accuracy, robustness, and ethical considerations.

RQ9 : What are the key future directions for advancing face swap detection research and applications?

A9 : The aim of this question is to identify emerging challenges, technological opportunities, and strategic research areas that can guide the development of more robust, generalizable, and ethically sound face swap detection systems.

IV. RESULTS AND DISCUSSION

This section presents the results and discussion based on the research questions guiding this survey. It provides a comprehensive analysis of the most commonly used datasets in face swap detection, the evaluation metrics for assessing model performance, and a comparative overview of various detection methods. Beyond performance analysis, this section also includes a detailed taxonomy of detection methods, an evaluation of cross-dataset generalization capabilities, and a dedicated exploration of recent innovative directions. Furthermore, it outlines best practices for building and deploying face swap detection tools responsibly and concludes with proposed future directions aimed at advancing the field both technically and ethically.

A. DATASETS

This section focuses on face swap datasets, which either exclusively feature face swapping as the primary deepfake technique or include it as a significant component of the dataset. The datasets presented here are specifically curated to provide a diverse range of face swapped videos, created using different state-of-the-art faceswap algorithms. These datasets serve as essential resources for research and development in deepfake detection, enabling the evaluation of models and techniques aimed at identifying face swapping manipulations in video content.

FaceForensics++ (FF++) is a widely recognized dataset in face swap detection research, containing over 1,000 original videos and four thousands of manipulated versions generated using four popular forgery techniques: DeepFakes, Face2Face, FaceSwap, and NeuralTextures. The dataset includes videos at different compression levels, such as raw videos without compression (C0), high-quality videos with a constant rate quantization parameter of 23 (C23), and low-quality videos with a constant rate quantization parameter of 40 (C40), to simulate real-world scenarios where video

A6 : The aim here is to explore the difficulties and shortcomings faced by current face swap detection models when applied to real-world scenarios.

RQ7 : What recent innovations have been proposed to improve the robustness, transparency, and deployability of face swap detection models?

TABLE 2. Summary of face swap detection datasets.

Dataset	FF++	Celeb-DF	DFDC	DFDC Preview	DF-TIMIT	Face Shifter	Deeper Forensics 1.0	UADFV
Released	2019	2019	2020	2019	2018	2019	2020	2018
Total videos	5000	6229	128,154	5250	960	2000	60000	98
Real content	1000	590	23654	1131	320	1000	50000	49
Fake content	4000	5639	104500	4119	640	1000	10000	49
Resolution	480p, 720p, 1080p	256x256	1080x1920	1080x1920	64x64 (LQ), 128x128 (HQ)	NA	1080x1920	294x500
Synthesis Methods	4	1	8	2	1	1	1	1
Forgery Techniques	DeepFakes, Face2Face, FaceSwap, Neural Textures	presented in the paper	DF-128, DF-256, MM/NN, NTH, FSGAN, StyleGAN, refinement, audio swaps	NA	faceswap-GAN	Face Shifter	DF-VAE	FakeApp
Participants Consent	No	No	Yes	Yes	No	No	Yes	No
# of Actors	NA	59	960	66	32	NA	100	49

quality can vary. FF++ excels in its comprehensive coverage of different manipulation techniques, making it invaluable for training and evaluating detection models. However, its reliance on specific forgery algorithms means that models trained on FF++ might not generalize well to novel or unseen manipulation methods [13].

The **FaceShifter dataset** is a specialized subset of the FaceForensics++ dataset, focusing on high-fidelity face swapping. It comprises 1,000 videos generated using the FaceShifter method, which employs a two-stage deep neural network to produce realistic and identity-preserving face swaps [14]. Its main strength lies in the visual fidelity of the manipulations, making it ideal for evaluating detection methods under challenging conditions. However, it is limited by its reliance on a single face swapping technique and lacks diversity in manipulation styles and real-world variability, which may impact the generalization of detection models.

The **Google DeepFake Detection (Google DFD) dataset** is a significant contribution to the field of synthetic media research, developed collaboratively by Google and Jigsaw. It comprises over 3,000 videos featuring both real and synthetically manipulated content, created using undisclosed deepfake generation techniques. The dataset includes videos of paid actors performing various actions, providing a diverse range of facial expressions, lighting conditions, and backgrounds. This diversity is crucial for training robust deepfake detection models. The DFD dataset has been integrated

into the FaceForensics++ benchmark suite, facilitating standardized evaluation of detection algorithms [15].

The **Celeb-DF** dataset is an excellent resource for studying face swap detection, as it contains over 5,600 deepfake videos featuring 590 original videos of 59 celebrities. This dataset was specifically designed to address the limitations of earlier deepfake datasets, offering high-quality manipulations with improved lip synchronization and fewer visible artifacts. The diverse range of celebrities, backgrounds, and video conditions make Celeb-DF particularly valuable for evaluating detection models that need to handle real-world variability. Its realistic and high-quality content provides a solid foundation for training algorithms that can detect subtle face swapping manipulations. This dataset's ability to produce highly convincing fakes ensures that models trained on Celeb-DF can generalize well to new, sophisticated face swap techniques [16].

The **DeepFake Detection Challenge (DFDC)** provides a massive dataset comprising approximately 23,654 real videos and 104,500 fake videos, totaling around 128,154 videos for 960 subjects, created to reflect real-world conditions with diverse faces, lighting environments, and manipulation techniques. DFDC includes both deepfake videos and original clips, emphasizing large-scale, high-quality content suitable for training deep learning models. Its extensive scale allows researchers to build robust models capable of handling varying video qualities and diverse conditions. However,

as with most datasets, its synthetic nature and biases in data generation can limit its applicability to real-world scenarios where manipulation techniques evolve rapidly [17].

A smaller subset of DFDC dataset, the **DFDC Preview dataset**, was released earlier to provide researchers with an initial resource for developing and testing deepfake detection models. This dataset consists of 5,214 videos featuring 66 individuals. While significantly smaller than the full DFDC dataset, it maintains diversity in facial expressions, lighting conditions, and manipulation techniques, making it useful for preliminary studies before scaling to larger datasets [18].

DeeperForensics 1.0 dataset is a large-scale dataset designed to advance real-world face forgery detection research. It comprises 60,000 videos totaling 17.6 million frames, making it one of the most extensive datasets in this domain. The dataset includes 50,000 real videos collected from 100 paid actors of diverse demographics, and 10,000 manipulated videos generated using a novel face swapping method called DeepFake Variational Auto-Encoder (DF-VAE). Its scale, diversity, and use of real-world compression make it valuable for training robust detection models. However, its focus on a single forgery method limits the evaluation of generalization across varied manipulation techniques [19].

The **DeepfakeTIMIT (DF-TIMIT) dataset** is a collection of 620 deepfake videos created using a GAN-based face swapping approach adapted from the open-source faceswap-GAN [20]. It was developed from the publicly available VidTIMIT database [21], where 16 visually similar pairs were manually selected, resulting in 32 subjects. For each subject, two deepfake models were trained: a low-quality version and a high-quality version. Since each individual in VidTIMIT has 10 video clips, this resulted in 320 deepfake videos per version. The dataset preserves the original audio tracks without any modifications, ensuring that only facial features are altered. DF-TIMIT provides a controlled dataset for evaluating deepfake detection techniques, offering both low- and high-resolution face swaps while maintaining speech authenticity [22].

The **WildDeepfake dataset** is a real-world benchmark designed to enhance the robustness of deepfake detection systems. WildDeepfake comprises 7,314 face sequences extracted from 707 deepfake videos sourced entirely from the internet. This approach introduces greater diversity in terms of scenes, lighting conditions, and manipulation techniques, thereby presenting a more challenging and realistic scenario for detection models. Its real-world origin makes it valuable for evaluating model generalization. However, it provides only image sequences rather than full videos, limiting temporal analysis, and its relatively small size may constrain training applications [23].

The **UADFV (University at Albany DeepFake Video) dataset** is one of the earliest publicly available datasets designed to support research in deepfake detection.

Introduced in 2018, it comprises 98 videos—49 real and 49 manipulated—generated using the FakeApp tool. The dataset primarily focuses on face swapping manipulations and was developed to facilitate the study of physiological signals, such as eye blinking, which are often inconsistent or absent in deepfake videos [24].

The **Div-DF dataset** is a collection of deepfake videos designed to support research in detecting various video manipulation techniques. It consists of 150 real videos featuring different celebrities and 250 deepfake videos, which are categorized into three types: 100 face swap videos, 100 facial reenactment videos, and 50 lip-sync videos [25]. The face swap deepfake videos are synthesized using the state-of-the-art face swap GAN (FSGAN) [26].

Table 2 presents a detailed comparison between the datasets used for face swap detection in specific. We can summarize that DFDC stands out as the largest dataset, with over 128,000 videos and 960 actors, providing high diversity and realism suitable for training and evaluating robust detection models. Meanwhile, Div-DF, DF-TIMIT, and UADFV are more limited in scale and resolution, which may constrain their effectiveness for generalization. Only a few datasets, such as DFDC, Google DFD, and DeeperForensics 1.0, explicitly mention obtaining participant consent, emphasizing ongoing ethical concerns in dataset construction.

Figure 3 presents the number of research studies that have utilized each of the datasets, showcasing the popularity and application of these resources in the field. FF++ leads the chart with 146 research studies, reflecting its widespread use in this field. The Celeb-DF dataset follows closely with 118 studies, indicating its importance for researchers focusing on celebrity-driven face swapping manipulations. DFDC is used in 73 studies, highlighting its significance for large-scale model development, while DFDC-Preview, with 15 studies, is notably less cited. This lower number likely reflects the smaller size of the DFDC-Preview dataset compared to the full version, limiting its appeal for large-scale model training. Datasets like Google DFD and FaceShifter are cited in 24 and 13 papers respectively. Google DFD offers high-quality, manually verified fake videos and contributes to developing more precise detectors, while FaceShifter is recognized for its state-of-the-art face swapping techniques, making it useful for evaluating detection methods against more challenging manipulations. Meanwhile, datasets like DeeperForensics 1.0, Wild Deepfake, DF-TIMIT, and UADFV are used less frequently, perhaps due to their limited scope, older generation techniques, or lack of recent updates. Lastly, Div-DF has been referenced in 1 study, indicating it is relatively new or not yet widely adopted. Overall, the figure highlights the prominence of FF++ and Celeb-DF in the research community, while also pointing out that the DFDC dataset is less frequently used compared to FF++ and Celeb-DF. This is likely due to the large size of the DFDC dataset,

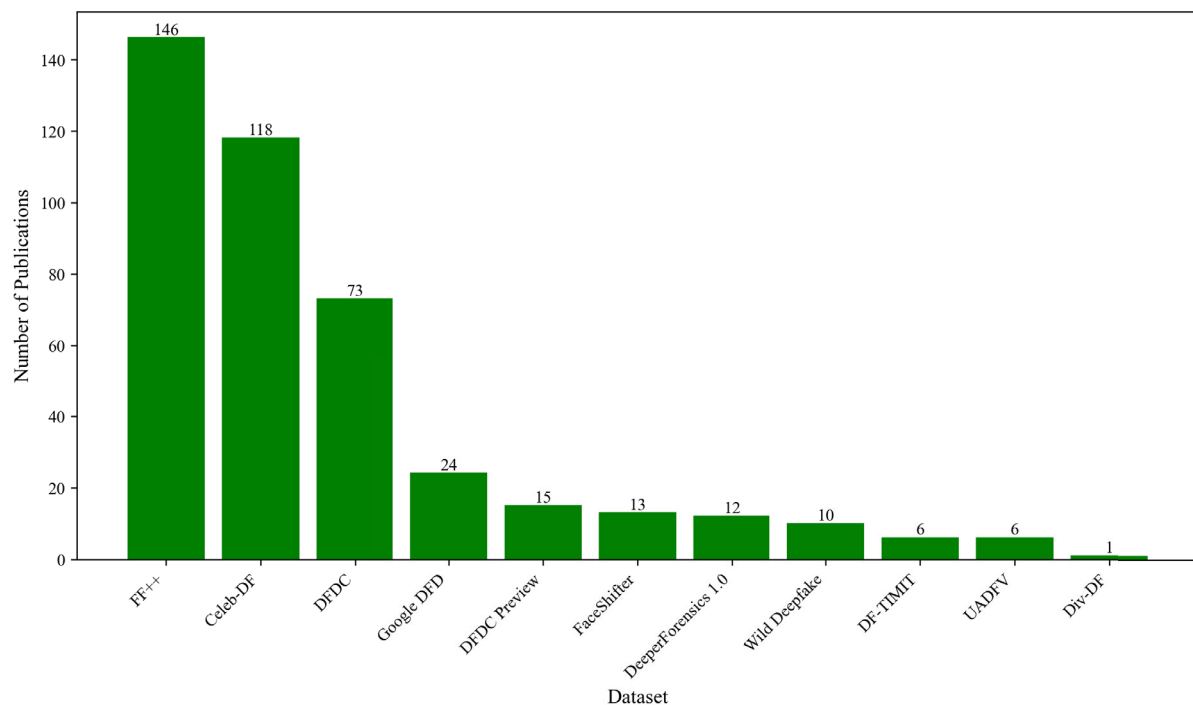


FIGURE 3. Usage of different datasets in research studies.

which requires significant computational resources, such as supercomputers, for efficient training and testing of deepfake detection models.

B. METHODS TAXONOMY

The rapid advancement of face swap technologies has led to significant research efforts aimed at developing robust detection methods, resulting in a broad range of techniques built on diverse computational foundations. To bring clarity to this evolving field, a taxonomy is presented to classify face swap detection models based on their core technical strategies. The taxonomy organizes methods into three principal categories: spatial methods, which analyze individual frames to identify visual or statistical inconsistencies; temporal methods, which examine patterns and discrepancies across sequences of frames; and spatiotemporal methods, which integrate spatial and temporal information to improve detection performance. Each primary category is further divided into subcategories reflecting specific methodological approaches, such as CNN-based architectures, attention mechanisms, frequency-domain analyses, and hybrid models that combine multiple techniques. This structured classification provides a comprehensive overview of the current landscape and highlights key trends and emerging directions in face swap detection research. Figure 4 presents a taxonomy of face swap detection methods, categorizing existing approaches into spatial, temporal, and spatiotemporal methods. The following explains the subcategories within each of these categories.

• Spatial Methods

The spatial category focuses on analyzing individual frames to detect face swaps. These methods examine visual and statistical inconsistencies within a single image, such as texture anomalies, color mismatches, or unnatural facial features.

– CNN-based Methods

These methods form a foundational subcategory within spatial face swap detection techniques. These approaches utilize classic Convolutional Neural Network (CNN) architectures such as ResNet, VGG, Xception, MobileNet, and EfficientNet to automatically learn and extract discriminative features from individual frames. Each of these models offers unique advantages: ResNet is known for its deep residual learning that helps prevent vanishing gradients, VGG emphasizes simplicity and depth, Xception employs depthwise separable convolutions for efficiency, MobileNet focuses on lightweight design for faster inference, and EfficientNet balances model size and accuracy through compound scaling.

– Attention-Enhanced Networks Methods

Attention-enhanced networks represent an advanced class of CNN-based methods that improve face swap detection by focusing on the most informative parts of an image. These models incorporate various types of attention mechanisms such as spatial, channel, multi-scale, region-based, or landmark attention to selectively highlight critical

features while suppressing irrelevant information. By directing the model's focus to key facial regions or important feature channels, attention mechanisms help capture subtle artifacts and inconsistencies introduced during face swapping. This targeted approach enhances detection accuracy and robustness compared to traditional CNNs that treat all image regions equally.

– Frequency/Domain Methods

These methods form another important subcategory within spatial face swap detection techniques. These approaches analyze images in transformed domains rather than relying solely on raw pixel information. Techniques such as Discrete Cosine Transform (DCT), Discrete Fourier Transform (DFT), high-frequency noise analysis, Local Binary Patterns (LBP), Gabor filters, and Spatial Rich Models (SRM) are commonly employed. By examining frequency components, textures, and fine-grained statistical patterns, these methods can detect subtle artifacts and inconsistencies that are often invisible in the spatial domain.

– Transformer Spatial Methods

They represent a growing subcategory in face swap detection, leveraging the power of transformer architectures to analyze spatial information within individual frames. Unlike traditional CNNs, transformers such as Vision Transformer (ViT) and Swin Transformer use self-attention mechanisms to capture long-range dependencies and global context across the entire image. These models process the input as sequences of patches, allowing them to identify subtle and widely distributed artifacts caused by face manipulation. When applied purely to spatial data, transformer-based models excel at modeling complex patterns and textures, offering a robust alternative to conventional approaches in detecting tampered content.

– Adversarial Network Methods

They leverage the power of adversarial learning to improve face swap detection by explicitly modeling the distribution differences between real and forged images. These approaches commonly employ GANs or adversarial discriminators to encourage the feature extractor to focus on intrinsic, manipulation-invariant features rather than superficial artifacts tied to specific forgery techniques or identities. By setting up a competitive learning framework between a generator and a discriminator, adversarial networks help the detection model become more robust against variations introduced by diverse forgery methods and improve its generalization to unseen manipulations.

– Ensemble & Hybrid Methods

Ensemble and hybrid methods form a versatile subcategory within spatial face swap detection.

These approaches combine multiple models, techniques, or learning paradigms to enhance detection performance and generalization. Ensemble methods aggregate predictions from different classifiers to improve robustness and reduce errors. Hybrid techniques integrate deep learning with traditional machine learning, for example, by extracting spatial features with CNNs and classifying them using classic algorithms like Support Vector Machine (SVM) or decision trees. Other strategies in this category include style transfer to detect visual inconsistencies, domain adaptation to improve cross-dataset performance, Siamese networks for comparing pairs of images, and meta-learning to quickly adapt to new manipulation techniques. Together, these diverse methods exploit the strengths of multiple approaches to address the complex challenges of face swap detection. Also, this category includes methods that combine multiple complementary cues or feature types such as biometric 3D facial shape and appearance information to improve face swap detection accuracy and robustness by leveraging the inconsistency between these heterogeneous features

• Temporal Methods

The temporal category focuses on detecting face swaps by analyzing inconsistencies across sequences of frames rather than single images. These methods leverage the temporal dynamics of facial movements, expressions, and transitions to uncover signs of manipulation that may not be evident in isolated frames. By studying patterns such as unnatural blinking, inconsistent head motion, or irregular lip synchronization, temporal approaches can identify anomalies introduced during the generation of fake videos. Among the studied papers, **Temporal Diversified Self-Contrastive Learning** is the only temporal method explored for face swap detection. This approach focuses on learning temporal consistency by contrasting sequences within videos to distinguish real motion patterns from manipulated ones. By encouraging diverse temporal representations, it becomes adept at capturing subtle inconsistencies across frames, such as unnatural facial movements or abrupt transitions, which are common indicators of face swap manipulations.

• Spatiotemporal Methods

Spatiotemporal methods form the third major category in face swap detection, integrating both spatial and temporal information to improve detection accuracy. These approaches analyze not only the visual details within individual frames but also the dynamics and consistency across sequences of frames. By combining spatial feature extraction with temporal modeling, spatiotemporal methods can detect subtle artifacts that manifest over time, such as inconsistent facial

movements or unnatural transitions. Techniques in this category often use hybrid architectures, merging CNNs with recurrent layers like Gated Recurrent Unit (GRU) or Long Short-Term Memory (LSTM), or employing 3D convolutional networks that jointly process spatial and temporal dimensions. This fusion enables a deeper understanding of video content, making spatiotemporal methods particularly powerful for identifying sophisticated face swap manipulations.

– 3D CNNs with Spatiotemporal Features Methods

They represent a core approach within the spatiotemporal category. Unlike traditional 2D CNNs that operate on individual frames, 3D CNNs extend convolution operations across both spatial and temporal dimensions, enabling the model to capture motion patterns and visual features simultaneously. By processing multiple consecutive frames as input, these networks can learn temporal dependencies and detect inconsistencies in facial movements or transitions that may indicate manipulation. This integrated analysis makes 3D CNNs particularly effective for video-based detection tasks, as they are capable of identifying both static artifacts and temporal anomalies introduced during face swapping.

– Hybrid CNN + RNN Methods

They are a significant subcategory within spatiotemporal face swap detection methods. In this approach, CNNs first extract spatial features from individual video frames, capturing details such as textures, edges, and facial landmarks. These spatial features are then fed into recurrent neural networks (RNNs), including architectures like LSTM, Bidirectional LSTM (Bi-LSTM), or GRU, which model the sequential relationships across frames. This combination allows the system to analyze both the appearance of each frame and the temporal dynamics of facial movements, helping detect subtle inconsistencies and unnatural transitions that signal face swaps.

– CNN + Transformer Hybrids Methods

They combine the strengths of CNNs and transformer architectures to process spatiotemporal inputs. In these models, CNNs typically handle spatial feature extraction from individual frames, capturing local textures and facial details. The extracted features are then passed to transformer modules, which use self-attention mechanisms to model long-range dependencies and temporal relationships across frames. This integration enables the network to effectively capture both fine-grained spatial artifacts and complex temporal patterns that may indicate manipulation. By leveraging CNNs' ability to learn local features and transformers' capacity for global context modeling, these hybrid

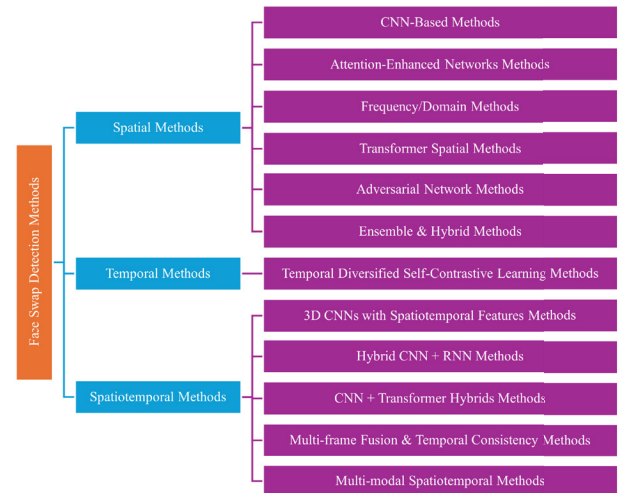


FIGURE 4. Taxonomy of face swap detection methods.

architectures offer enhanced detection performance in challenging video-based scenarios.

– Multi-frame Fusion & Temporal Consistency Methods

The Multi-frame Fusion and Temporal Consistency subcategory focuses on methods that emphasize the relationships and consistency between consecutive frames in a video. These techniques analyze frame-to-frame coherence, motion patterns, and temporal clues to detect anomalies. By fusing temporal information with spatial features, models can identify inconsistencies such as abrupt changes in facial appearance, unnatural movements, or irregular blinking that are difficult to detect when examining frames in isolation. Approaches in this category often employ fusion strategies to combine data from multiple frames, enhancing the robustness and accuracy of detection by leveraging both spatial details and temporal dynamics.

– Multi-modal Spatiotemporal Methods

Multi-modal spatiotemporal approaches integrate diverse types of information such as spatial, temporal, frequency-domain features, and occasionally depth or geometric data to enhance face swap detection. By combining these multiple modalities, these methods capture a richer and more comprehensive representation of the facial content and its dynamics. For example, frequency features can reveal subtle artifacts invisible in the spatial domain, while depth or geometric cues help detect inconsistencies in 3D facial structure. Integrating temporal information further allows these models to track changes and anomalies across frames.

The distribution of papers across the proposed taxonomy of face swap detection methods reveals clear research trends and areas of focus within the field as shown in Table 3.

TABLE 3. Distribution of papers within the taxonomy of face swap detection methods.

Main Category	Subcategory	# of papers	Percentage	Papers' References
Spatial Methods	CNN-Based Methods	46	28.57%	[27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72]
	Attention-Enhanced Networks Methods	32	19.88%	[73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [83], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103]
	Frequency/Domain Methods	19	11.8%	[104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122]
	Transformer Spatial Methods	9	5.59%	[123], [124], [125], [126], [127], [128], [129], [130], [131]
	Adversarial Network Methods	4	2.48%	[132], [133], [134], [43]
	Ensemble & Hybrid Methods	16	9.94%	[135], [136], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149], [150]
Temporal Methods	Temporal Diversified Self-Contrastive Learning Methods	1	0.62%	[151]
Spatiotemporal Methods	3D CNNs with Spatiotemporal Features Methods	4	2.48%	[152], [153], [154], [155]
	Hybrid CNN + RNN Methods	15	9.32%	[156], [157], [158], [159], [160], [161], [162], [163], [164], [165], [166], [167], [168], [169], [170]
	CNN + Transformer Hybrids Methods	3	1.86%	[171], [172], [173]
	Multi-frame Fusion & Temporal Consistency Methods	7	4.35%	[174], [175], [176], [177], [178], [179], [180]
	Multi-modal Spatiotemporal Methods	5	3.11%	[181], [182], [183], [184], [185]

Spatial methods dominate the landscape, accounting for approximately 78% of the surveyed works. Among these, CNN-based approaches are the most prevalent, representing around 28.5% of all papers, reflecting the strong reliance on convolutional architectures for spatial feature learning in face swap detection. Attention-enhanced networks follow, comprising about 19.8%, indicating the growing interest in mechanisms that help models focus on critical regions of manipulated images. Frequency and domain-based methods make up roughly 11.8%, suggesting a significant subset of researchers are exploring the frequency domain to capture subtle manipulation artifacts. Transformer spatial methods and adversarial networks remain smaller niches within spatial approaches, while ensemble and hybrid methods demonstrate moderate popularity, highlighting attempts to improve robustness by combining multiple techniques.

Temporal methods are minimally represented, with only a single paper (0.6%), suggesting that purely temporal approaches are less explored in isolation. In contrast, spatiotemporal methods collectively account for over 21%

of the literature, reflecting increased attention on capturing temporal consistency and multi-frame correlations to detect manipulations across video sequences. Within spatiotemporal methods, hybrid CNN-RNN architectures are prominent (about 9.3%), underlining the synergy between spatial and sequential modeling. Multi-frame fusion and multi-modal strategies also appear in notable proportions, emphasizing the community's effort to leverage both temporal and cross-modal cues for more reliable detection. Overall, the taxonomy indicates that while spatial cues remain foundational, there is a significant and growing shift towards exploiting spatiotemporal information and advanced network designs to address the challenges of increasingly sophisticated face swap techniques.

C. PERFORMANCE MEASURES

As the accuracy and reliability of deepfake detection models are critical in real-world applications, selecting the appropriate evaluation metrics is essential. In this context, a variety of performance measures have been employed to

assess the effectiveness of face swap detection models. These metrics help evaluate different aspects of model performance, such as discrimination ability, error rates, and the balance between false positives and false negatives. The section discusses the most commonly used metrics in face swap detection studies, including area under the curve, accuracy, F1 score, recall, precision, and others, offering insights into the evaluation landscape of face swap detection technologies.

Area Under the Curve (AUC) is a key performance metric in binary classification, measuring the ability of a model to distinguish between two classes. It is derived from the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR). AUC values range from 0 to 1, with 1 indicating perfect classification and 0.5 suggesting random guessing.

Partial Area Under the Curve (pAUC) is a performance metric derived from the ROC curve, used to evaluate the effectiveness of a classifier over a specific range of the FPR. While the standard AUC calculates the area under the entire ROC curve, pAUC focuses on a specific portion of the curve, usually corresponding to a particular range of FPR values.

Accuracy (ACC) is a performance metric that measures the overall correctness of a model's predictions. It is calculated as the ratio of correct predictions (both true positives and true negatives) to the total number of predictions, as shown in Equation (1).

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Samples}} \quad (1)$$

Precision is a performance metric that measures the accuracy of the positive predictions made by the model. It can be calculated using Equation (2). In other words, precision answers the question: "Of all the instances the model classified as positive, how many were actually positive?"

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

Recall (also known as **Sensitivity** or **TPR** or **True Detection Rate (TDR)**) is a performance metric that measures how well the model identifies positive instances. It can be calculated using Equation (3). Recall answers the question: "Of all the actual positive instances, how many did the model correctly identify?"

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3)$$

The **False Negative Rate (FNR)** is representing the proportion of actual positives that are incorrectly classified as negatives. It is essentially the complement of Recall (TPR). It can be calculated using Equation (4).

$$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}} \quad (4)$$

The **F1 Score** is a performance metric that combines both precision and recall into a single value, providing a balance between the two. It is particularly useful when the class distribution is imbalanced, as it takes both false positives and

false negatives into account. The F1 Score is the harmonic mean of precision and recall, which means that it gives more weight to lower values. It can be calculated using Equation (5).

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

Equal Error Rate (EER) represents the point at which the False Acceptance Rate (FAR) and the False Rejection Rate (FRR) are equal. FAR is the proportion of negative samples incorrectly classified as positive (false positives). FRR is the proportion of positive samples incorrectly classified as negative (false negatives). The EER is the point where FAR equals FRR, and it is often used as a threshold for determining the trade-off between these two errors. The lower the EER, the better the model, as it indicates a better balance between false positives and false negatives.

Half Total Error Rate (HTER) is defined as the average of the FAR and FRR. HTER provides a balanced view of both types of errors by treating them equally, which can be useful when it is important to minimize both false positives and false negatives simultaneously. It can be calculated using Equation (6).

$$\text{HTER} = \frac{1}{2} (\text{FAR} + \text{FRR}) \quad (6)$$

Matthews Correlation Coefficient (MCC) is a performance metric for binary classification that takes into account True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). It is considered a balanced measure, even when the classes are imbalanced. The MCC provides a value between -1 and 1 . 1 represents a perfect prediction, 0 indicates a random prediction (no better than random chance), -1 indicates total disagreement between prediction and actual values. It can be calculated using Equation (7).

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (7)$$

The **True Negative Rate (TNR)**, also known as **Specificity**, is a performance metric for binary classification. It measures the proportion of actual negatives that are correctly classified as negative. It is the complement of the FPR. It can be calculated using Equation (8).

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (8)$$

In analyzing the performance measures used across various research studies, it is clear from Figure 5 that AUC and ACC are the most commonly employed metrics, appearing in 137 and 124 studies, respectively. These measures are crucial for evaluating the overall effectiveness of deepfake detection models, with AUC assessing a model's ability to distinguish between real and fake faces, and accuracy offering a straightforward evaluation of correct predictions. Measures such as F1 Score (21 studies) and Recall (22 studies) are also

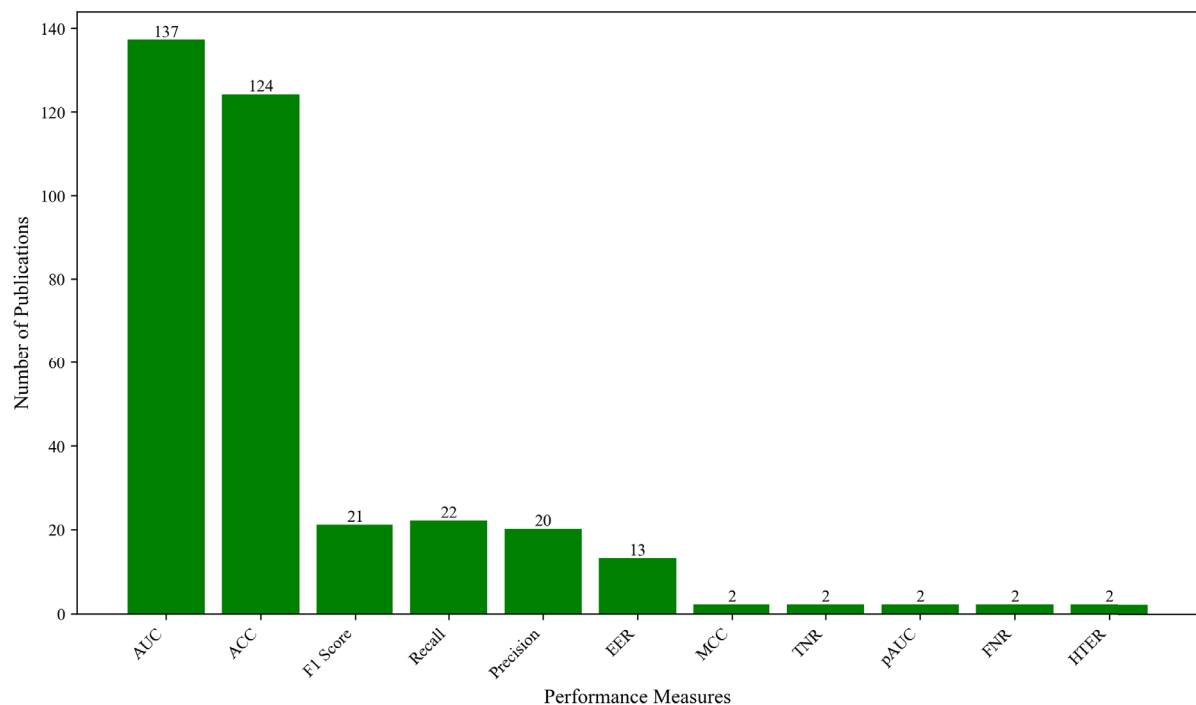


FIGURE 5. Usage of different performance measures in research studies.

commonly used, particularly when dealing with imbalanced datasets where the cost of false negatives (i.e., misclassifying a fake face as real) is significant. Precision (20 studies) complements these by evaluating the model's ability to avoid false positives (i.e., incorrectly classifying a real face as fake). EER (13 studies) is an important metric in face swap detection, as it helps balance the trade-off between false positive and false negative rates, which is critical for applications requiring high confidence in detection. Metrics like MCC, TNR, pAUC, FNR, and HTER (each with 2 occurrences) are less commonly used but still significant in specific contexts, such as assessing true negative rates, partial AUC, or overall error rates. This distribution of measures suggests that while AUC and ACC dominate the research landscape in face swap detection, a smaller subset of studies integrates more specialized metrics to address particular challenges, such as improving detection accuracy for hard-to-detect manipulations or minimizing false positives and negatives in real-world scenarios.

D. PERFORMANCE MEASURE VALUES DISCUSSION

In the context of performance measure values, the accuracy and AUC results presented in Figures 6 and 7 provide insightful comparisons across three different datasets: Celeb-DF, FF++, and DFDC. In Figure 7, the accuracy data highlights that the majority of models on Celeb-DF and FF++ datasets perform in the higher accuracy range (90-100), with Celeb-DF having 19 models in this category and FF++ having 13. However, for the DFDC dataset, the

number of models within this range is significantly lower (7), suggesting that the models might be less effective on this dataset. A possible reason for this discrepancy could be the higher complexity or quality of the data in DFDC, which might make it harder for existing models to achieve higher accuracy, indicating the need for more robust or specialized approaches for DFDC.

Moving to the lower accuracy ranges, it is noteworthy that Celeb-DF shows a greater number of models (<70 accuracy) compared to FF++ and DFDC. This might indicate that, despite being one of the more commonly used datasets for face swap detection, Celeb-DF could present more challenging or diverse scenarios that contribute to lower model performance. FF++, on the other hand, has relatively fewer models in the lower accuracy ranges, suggesting that the dataset may offer features that are easier for face swap detection models to handle, although it still contains some cases where models fail to reach high performance.

When examining the AUC results as shown in Figure 6, a similar trend emerges. Celeb-DF and FF++ both have a significant number of models within the 90-100 AUC range, with 28 and 31 models, respectively. This indicates that these datasets not only enable high accuracy but also lead to better discrimination of face swap detection tasks. AUC, being a metric that evaluates model performance across all possible classification thresholds, gives further confidence that these models, especially those trained on Celeb-DF and FF++, are highly reliable in distinguishing between real and swapped faces. DFDC, however, shows a much lower AUC in the

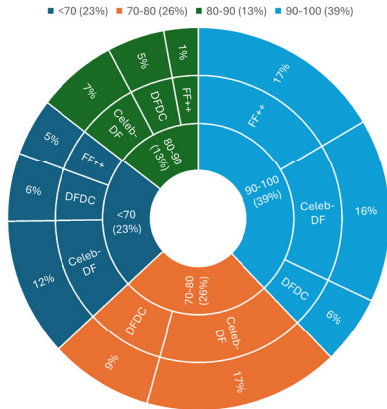


FIGURE 6. Distribution of performance measure values based on AUC.

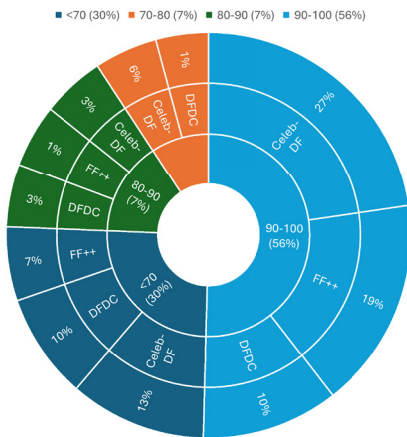


FIGURE 7. Distribution of performance measure values based on Accuracy.

90-100 range (10 models), reflecting a potentially higher difficulty in achieving strong classification results on this dataset.

In the middle and lower AUC ranges (80-90 and 70-80), Celeb-DF maintains a higher number of models than FF++ or DFDC, particularly in the 70-80 range, where 30 models fall. This suggests that Celeb-DF, while yielding high accuracy, may require more nuanced model calibration to improve the AUC, indicating some challenges in optimizing models for both high accuracy and high discrimination power. DFDC, in contrast, has fewer models in these intermediate ranges, which could imply that the dataset either requires more advanced techniques for performance improvement or that the current methods have not been sufficiently optimized for its particular characteristics.

The analysis of the top 10 research papers based on AUC and accuracy for face swap detection across the FF++, Celeb-DF, and DFDC datasets reveals considerable variation in the performance of the models. This discrepancy highlights both the advances and challenges in this field of research.

For the FF++ dataset, the AUC of the top 10 papers ranges from 99.94 to 99.8 as shown in Table 4, demonstrating that

TABLE 4. Top 10 research papers based on AUC for FF++ dataset.

Reference	Year	Publisher	AUC
[139]	2024	SinceDirect	99.94
[159]	2022	ACM	99.94
[168]	2022	ACM	99.93
[178]	2024	Springer	99.9
[75]	2023	IEEE	99.89
[177]	2022	Springer	99.85
[143]	2022	SinceDirect	99.84
[94]	2023	Springer	99.83
[114]	2024	SinceDirect	99.82
[144]	2023	ACM	99.8

TABLE 5. Top 10 research papers based on Accuracy for FF++ dataset.

Reference	Year	Publisher	Accuracy
[41]	2021	ACM	99.79
[170]	2024	Springer	99.78
[165]	2022	Springer	99.2
[79]	2024	SinceDirect	99.01
[173]	2024	Springer	99
[100]	2024	SinceDirect	98.05
[139]	2024	SinceDirect	98
[142]	2023	Springer	98
[148]	2023	Springer	97.7
[130]	2023	Springer	96.75

most models achieve very high performance in distinguishing between real and swapped faces. The highest AUC, [159], is 99.94, indicating that the model is exceptionally proficient in detecting face swaps with near-perfect discriminatory ability. However, the lowest AUC in this dataset, [166], is 65.47, which suggests that certain models are significantly less effective, potentially due to limitations in the dataset or the complexity of the face swap scenarios, or the model is not generalized well. Regarding accuracy, the top 10 papers range from 99.79 to 96.75 as shown in Table 5, with [41] achieving 99.79, signaling excellent detection capabilities. In contrast, [38], with an accuracy of 88.1, reflects a notable drop in performance, indicating that some models are prone to false positives or other errors in detection.

For the Celeb-DF dataset, the AUC of the top 10 papers ranges from 99.98 to 99.74 as shown in Table 6, with [55] achieving the highest AUC of 99.98. This reflects the fact that some models in this dataset can almost flawlessly detect face swaps. However, [166], with an AUC of 65.61, shows that there is still a significant gap in model performance, possibly due to the variety of face manipulations and the difficulty of distinguishing swapped faces in this particular dataset. The accuracy for the top 10 papers in Celeb-DF ranges from 99.93 to 97.74 as shown in Table 7, with [157] at 99.93, indicating strong performance. However, [82], at 63.27, is a stark reminder that some models are not able to achieve consistently high accuracy, particularly in real-world scenarios where face swaps may be more subtle or harder to detect.

For the DFDC dataset, the AUC ranges from 99.84 to 95.7 as shown in Table 8. The highest AUC, [92], is 99.84,

TABLE 6. Top 10 research papers based on AUC for Celeb-DF dataset.

Reference	Year	Publisher	AUC
[55]	2024	IEEE	99.98
[31]	2023	IEEE	99.94
[103]	2022	IEEE	99.94
[64]	2024	Springer	99.93
[157]	2022	SinceDirect	99.91
[81]	2023	SinceDirect	99.87
[79]	2024	SinceDirect	99.86
[38]	2021	ACM	99.79
[110]	2021	IEEE	99.78
[77]	2024	SinceDirect	99.74

TABLE 7. Top 10 research papers based on Accuracy for Celeb-DF dataset.

Reference	Year	Publisher	Accuracy
[157]	2022	SinceDirect	99.93
[175]	2021	ACM	99.78
[64]	2024	Springer	99.67
[55]	2024	IEEE	99.65
[77]	2024	SinceDirect	99.11
[31]	2023	IEEE	98.73
[79]	2024	SinceDirect	98.65
[103]	2022	IEEE	98.59
[80]	2023	Springer	97.81
[122]	2023	Springer	97.47

TABLE 8. Top 10 research papers based on AUC for DFDC dataset.

Reference	Year	Publisher	AUC
[92]	2023	Springer	99.84
[66]	2024	Springer	99.63
[32]	2023	IEEE	99.52
[182]	2023	ACM	99.2
[84]	2024	IEEE	98.79
[99]	2023	SinceDirect	98.27
[125]	2023	Springer	97.9
[131]	2022	Springer	97.8
[77]	2024	SinceDirect	97.37
[109]	2023	IEEE	95.7

which shows that some models achieve outstanding results in detecting face swaps. On the other hand, [74], with a value of 64.15, emphasizes the considerable disparity in model efficacy, where some models are less capable of distinguishing face swaps with high certainty. The accuracy in DFDC ranges from 99.91 to 81.65 as shown in Table 9, with [157] achieving 99.91, which indicates that the best models in this dataset perform at near-perfect levels. However, [82], at 65.76, highlights that a few models struggle to consistently detect face swaps, possibly due to the complexity of the dataset or the techniques used.

Overall, these results underscore the significant variability in model performance across both AUC and accuracy metrics. While some research papers achieve near-perfect results in face swap detection, others lag behind due to factors such as the diversity of manipulations in the datasets, the algorithms used, and the inherent challenges in detecting face swaps.

The following research papers have been featured in three or more of the top-performing studies in face swap detection research: [31], [77], [79], [100], [130], [139], and [157]. This

TABLE 9. Top 10 research papers based on Accuracy for DFDC dataset.

Reference	Year	Publisher	Accuracy
[157]	2022	SinceDirect	99.91
[92]	2023	Springer	98.39
[100]	2024	SinceDirect	97.42
[170]	2024	Springer	92.72
[77]	2024	SinceDirect	92.39
[130]	2023	Springer	92.16
[41]	2021	ACM	91.69
[175]	2021	ACM	89.8
[109]	2023	IEEE	88.78
[31]	2023	IEEE	81.65

suggests that these research papers hold significant potential in the field of face swap detection.

E. CROSS-DATASET GENERALIZATION PERFORMANCE

Evaluating the generalization ability of face swap detection models is essential for understanding their robustness in real-world scenarios, where the distribution of test data often differs significantly from the training set. While in-domain evaluations may report high performance, they do not reliably capture how models behave when exposed to unfamiliar manipulations or acquisition conditions. Cross-dataset evaluation provides a more realistic assessment by measuring performance when models are tested on datasets that were not seen during training.

The papers included in this generalization analysis represent the top-performing methods in the field, as highlighted in the previous section. These studies were selected based on their state-of-the-art performance when evaluated within the same dataset. By analyzing how these leading models perform in cross-dataset scenarios, the aim is to assess their ability to generalize beyond the data distributions they were originally trained on.

The results presented in Table 10 show substantial variability across dataset pairs, reflecting the challenges of domain shift in deepfake detection. The table presents the AUC results reported by these papers, as AUC was the most consistently used metric across studies. Since only a few papers reported accuracy, AUC was adopted as the primary basis for comparison in evaluating cross-dataset generalization performance. In the table, notation such as (FF++ → CelebDF) indicates that the model was trained on the FF++ dataset and tested on the CelebDF dataset. This convention applies similarly to all other dataset pairs shown.

As illustrated in the table, the majority of studies have focused on training their models using the FF++ dataset and evaluating performance on the CelebDF dataset. The results across these studies range widely, from 54 to 95, but most values fall within the 70s. This indicates a noticeable drop in performance when models are tested on a different dataset, compared to their performance on the same dataset, where results commonly exceed 90. Such a gap highlights the generalization challenge faced by many face forgery

detection models, suggesting that they are often overfitted to dataset-specific characteristics.

A similar trend is evident in models trained on FF++ and tested on the DFDC dataset. Again, the performance declines significantly in cross-dataset evaluations, with most results clustering in the 70s range. This consistent degradation across both CelebDF and DFDC confirms that current models struggle with domain shifts and are limited in their ability to generalize effectively to unseen data distributions.

For the less common scenarios, where models are trained on CelebDF and tested on FF++ or DFDC, or trained on DFDC and tested on FF++ or CelebDF, only a limited number of studies have explored these directions. Due to the small sample size, it is difficult to draw strong conclusions. However, the available results still follow a similar pattern, with most scores falling within the 70s, reinforcing the observation that generalization remains a persistent challenge regardless of the training dataset.

It is important to note that the results do not provide sufficient evidence to conclude that training a model on a specific dataset will ensure strong generalization to others. The performance varies significantly depending on the source and target datasets, indicating that success on one dataset does not guarantee comparable results on another. This variability underscores the challenges of generalization and highlights the need for more robust and domain-adaptive approaches in face swap detection.

Even when considering models that incorporate self-supervised pre-training, generalization performance remains within a similar range. For instance, as demonstrated in [184], a self-supervised pre-trained model tested on the DFDC dataset achieved an AUC of 78.02. This value aligns with the broader trend observed across most models evaluated in cross-dataset settings, where AUC scores commonly fall within the 70s. This indicates that, despite the promise of self-supervised methods in improving feature representation, they have not yet achieved a significant leap in cross-dataset performance, reinforcing the persistence of domain shift issues in face swap detection.

In summary, the generalization analysis reveals a consistent drop in performance when models are evaluated across different datasets, despite achieving high accuracy within the same dataset. Most cross-dataset results fall within the 70s range, reflecting the models' limited ability to adapt to new data distributions. While the FF++ to CelebDF and DFDC scenarios are well-studied, other training-testing combinations remain underexplored. The findings emphasize the ongoing need for developing more robust, dataset-agnostic methods capable of maintaining high performance across diverse and unseen environments.

F. CHALLENGES AND LIMITATIONS

Despite significant progress in face swap detection technologies, the field continues to face a range of critical challenges and limitations that hinder the reliability, generalizability, and real-world deployment of detection systems. As deepfake

generation methods become increasingly sophisticated and accessible, detection models must contend with evolving threats, diverse content sources, and operational constraints. From variability in image and video quality to adversarial attacks, privacy and legal issues, imbalanced datasets, generalization to unseen data, these challenges highlight the complex and dynamic nature of face swap detection.

• Variability in Image and Video Quality

Variability in image and video quality remains one of the most critical challenges in face swap detection, particularly when moving from well-structured datasets to uncontrolled, real-world scenarios. While datasets like FF++, Celeb-DF, DFDC, DFDC Preview, DeepFakeTIMIT, DIV-DF, FaceShifter, Google DFD, DeeperForensics 1.0, WildDeepfake, and UADFV provide a strong foundation for training and evaluating detection models, they each exhibit limitations in capturing the full spectrum of quality degradations present in user-generated and socially shared content. FF++ provides multiple forgery methods and includes both high-quality and heavily compressed video versions. Still, models trained on it often fail to generalize to real-world media with more complex degradations. Celeb-DF improves realism by offering more natural facial expressions and lighting conditions, but it lacks extensive environmental diversity and compression variability.

DFDC and DFDC Preview aim to address these gaps with broader actor diversity, varied lighting, occlusion, and compression. Despite this, DFDC-trained models still encounter generalization issues when applied to social media content characterized by erratic quality and editing artifacts.

DeepFakeTIMIT offers both low- and high-quality video pairs, but these are generated under highly controlled recording conditions, limiting their usefulness in modeling natural video noise or platform-specific compression. Similarly, DIV-DF includes diverse manipulation methods but doesn't adequately simulate the wide range of visual artifacts seen in the wild.

FaceShifter and Google DFD introduce high-fidelity face swaps and improve photorealism, yet they focus on aesthetic quality and do not account for the degradations caused by everyday usage scenarios such as smartphone recordings or reuploads. DeeperForensics 1.0 stands out by explicitly modeling real-world distortions like motion blur, noise, and compression, making it one of the more realistic datasets for quality variance.

On the other end of the spectrum, WildDeepfake and UADFV are sourced from the internet, offering naturally occurring degradations. However, they are limited in size and manipulation diversity, reducing their overall training effectiveness for deep learning models.

In practical terms, detection models trained on these datasets may demonstrate strong performance under benchmark conditions but struggle with generalization

TABLE 10. Cross-Dataset AUC Performance Across FF++, CelebDF, and DFDC.

Reference	FF++ → CelebDF	FF++ → DFDC	CelebDF → FF++	CelebDF → DFDC	DFDC → FF++	DFDC → CelebDF
[99]	69.5					
[114]	73.65					
[84]	60.06	64.36	59.78	60.16		
[142]	87.5		76.72			
[92]	70.1					
[75]		73.82				
[64]	95.76					
[125]	75.1					
[170]		92.0				
[77]	77					
[159]	87.72	73.24	88.69	77.35	77.91	84.22
[109]	79.3					
[103]	68.71	69.06				
[66]						63.42
[94]		79.6				
[31]		79.0				
[177]	77.73					
[38]	71.93		64.47			
[168]	87.72	73.24	88.69	77.35	77.91	84.22
[32]	54.87	62.73	73.96	72.71	70.39	80.24
[144]	86.0					
[143]	71.61					
[178]	78.47					
[175]	75.58					

due to quality mismatches. Therefore, variability in quality remains a critical barrier to deploying deepfake detection systems effectively in real-world environments.

• Data Imbalance

Data imbalance is a significant challenge in the face swap detection field, where the number of fake samples often greatly exceeds or falls short of real samples depending on the dataset. This imbalance can lead to biased models that either overfit to the dominant class or underperform in real-world detection scenarios where both real and fake content coexist in varied proportions. For example, FF++ contains 1000 real and 4000 fake videos, heavily favoring manipulated content. This skew can cause detection models to learn patterns more strongly associated with fakes, while underrepresenting authentic content characteristics. Similarly, Celeb-DF presents 590 real versus 5639 fake videos, reinforcing this disproportion. Although this may seem useful for training robust classifiers, it risks promoting models that show inflated performance on synthetic benchmarks but fail under balanced or real-world distributions.

DFDC offers a broader dataset with 23,654 real and 104,500 fake videos, aiming for diversity in manipulations and content. Despite this scale, the extreme imbalance in class distribution can still affect model calibration, especially in scenarios where equal precision and recall are critical. DFDC Preview, a smaller subset, also reflects similar class imbalance.

In DeepFakeTIMIT, only 320 real and 640 fake videos are provided, further demonstrating an overrepresentation

of synthetic content. DIV-DF and FaceShifter both include fewer real samples, while maintaining a higher number of fake ones, introducing risks of biased learning.

On the other hand, WildDeepfake and UADFV, although smaller, are relatively balanced, each containing 49 real and 49 fake instances. However, their limited size restricts their use for deep learning training and evaluation purposes.

Some datasets, such as Google DFD and DeeperForensics 1.0, also suffer from data imbalance, albeit in different ways. Google DFD includes 363 real and 3000 fake videos, showing a significant bias toward fake content. Conversely, DeeperForensics 1.0 presents the opposite scenario, with 50,000 real and only 10,000 fake videos, resulting in a strong bias toward authentic content. In both cases, the imbalance can skew the learning process: models trained on Google DFD may overfit to fake artifacts, while those trained on DeeperForensics 1.0 may struggle to detect manipulated content due to underexposure to it. These inconsistencies further highlight the difficulty of building generalizable face swap detectors when the ratio of real to fake content varies widely between datasets.

• Generalization to Unseen Data

A critical challenge in face swap detection lies in achieving robust generalization to unseen data—particularly when detection models are deployed outside the conditions of their training datasets. While many methods show impressive performance on in-distribution test sets, they often fail to maintain accuracy when evaluated

on deepfakes generated using different techniques, sources, or environmental conditions.

The analysis shown in Section IV-E of cross-dataset results reveals that generalization remains a significant challenge in face swap detection. While many models achieve high performance when trained and tested on the same dataset, their effectiveness drops notably when applied to unseen datasets. Most cross-dataset AUC scores fall in the 70s, in contrast to within-dataset scores that often exceed 90. Only a limited number of studies have explicitly evaluated their models across diverse datasets, and among those, few maintain consistently high performance.

Several factors contribute to this issue:

Overfitting to dataset-specific patterns: Many models inadvertently learn superficial cues (e.g., compression artifacts or generation tool fingerprints) specific to datasets, which do not persist in other sources.

Rapid evolution of face swap methods: New manipulation algorithms continue to emerge, producing more realistic results with fewer detectable artifacts. Models trained on older deepfake styles often fail to detect these improved forgeries.

Domain shift between curated datasets and real-world media: Datasets often lack the variability found in uncontrolled online videos, such as varying lighting, occlusions, camera motion, and heavy compression. This leads to performance degradation when models are deployed in real scenarios.

Limited dataset diversity: As mentioned above the existing datasets aim to include a broader range of manipulations, but they still fall short in capturing the full spectrum of manipulation styles, demographics, and media quality seen in the wild.

• Adversarial Manipulations

As face swap detection models become more advanced, so do the adversarial techniques designed to evade them. Adversarial manipulations pose a serious and growing threat to the reliability of deepfake detection systems, especially in security-critical applications such as misinformation prevention, identity verification, and digital forensics.

These manipulations involve the intentional crafting of content—either during or after the face swapping process—to fool machine learning models while maintaining perceptual realism to the human eye. This includes techniques such as:

Adversarial perturbations: Tiny, imperceptible changes to pixel values can be applied to face swapped images or frames, causing deep learning-based detectors to misclassify them as real.

Post-processing attacks: Applying transformations such as blurring, noise addition, compression artifacts, or color grading can mask the telltale signs of manipulation that detectors rely on.

GAN-based adversarial training: More sophisticated attacks now use GANs not only to create the face swap but also to train against known detection models, effectively “learning” how to bypass them.

Detector-aware attacks: In white-box scenarios, attackers may have access to the target model’s structure and parameters, allowing them to craft highly targeted adversarial examples that maximize misclassification.

The challenge is further amplified by the fact that most benchmark datasets do not contain adversarial examples, meaning that models trained on these datasets are often not robust to such attacks. Additionally, as new detection models are published, attackers continuously adapt, creating an ongoing arms race between detection and evasion.

• Privacy and legal issues

As face swap and deepfake detection technologies evolve, they raise significant privacy, ethical, and legal concerns. These issues impact not only how datasets are collected and models are trained, but also how detection systems are deployed and used in the real world.

Most datasets use real videos of people, including public figures, actors, and volunteers. The use of these identities, especially when altered to create manipulated (fake) videos, raises serious concerns.

- 1) **Consent:** Whether individuals explicitly agreed for their images/videos to be used in synthetic manipulation.
- 2) **Right to be forgotten:** In regions like the EU (under General Data Protection Regulation (GDPR)), individuals have the right to request removal of personal data, which complicates dataset retention and sharing.
- 3) **Exploitation:** Use of likenesses without permission, especially when content is manipulated, can be ethically problematic.

FF++, Celeb-DF, UADFV, Div-DF, FaceShifter, and WildDeepfake use publicly available YouTube videos of celebrities, often without explicit consent. DF-TIMIT is based on the TIMIT dataset and may have consent for voice data but not necessarily for facial manipulation.

DFDC and DFDC Preview obtained videos from paid actors, with more structured consent, making them ethically safer. Google DFD and DeeperForensics-1.0 claim to have used explicit volunteer consent.

Also, many datasets do not offer mechanisms for individuals to remove their data or even be informed of its use. This lack of data control could make some datasets non-compliant with privacy laws, especially if used in commercial systems.

G. INNOVATIVE DIRECTIONS

Face swap detection has evolved rapidly in response to increasingly sophisticated manipulation techniques and growing real-world concerns. While much of the research has

focused on improving detection accuracy, recent efforts have shifted toward enhancing model robustness, transparency, efficiency, and ethical deployment. This section highlights key innovative directions in the field, including adversarial defense strategies, explainability techniques, lightweight models for edge deployment, and privacy-preserving training.

• Adversarial Defense Strategies

With the growing deployment of face swap detectors in high-stakes domains such as social media content moderation and digital forensics, ensuring robustness against adversarial conditions has become essential. Adversarial perturbations whether crafted deliberately or introduced through common post-processing operations like noise, compression, or blurring, can significantly degrade the reliability of detection systems. This vulnerability underscores the need for adversarial defense strategies that enhance model resilience under real-world distortions.

One prominent example addressing this challenge is the model proposed in [173], which integrates both architectural and feature-level defense mechanisms. The authors introduce a hybrid system combining ResNet-18 with cross-attention ViT transformers, enhanced by a Global Texture Module that extracts Gram matrix-based texture representations and LBP at each ResNet down-sampling stage. This design facilitates the extraction of both fine-grained local details and broader contextual features, improving the model's capacity to withstand adversarial perturbations.

In their robustness evaluation, the model was tested under multiple unseen corruptions, including Gaussian noise, contrast shifts, pixelation, and compression artifacts (H.264). Results revealed significant performance degradation in conventional models such as MesoNet, CviT, and CNN-Net, with accuracy falling as low as 50% under noise. In contrast, the LBP-ViT model consistently maintained superior robustness, achieving up to 98% accuracy on the DFDC Preview dataset, even when other models failed to reach 80%. This illustrates the effectiveness of multi-scale feature extraction and attention-driven architectures in defending against input-level adversarial distortions.

Moreover, the study suggests two key strategies for enhancing adversarial robustness in face swap detection: (1) employing advanced data augmentation during training to simulate real-world corruptions and (2) adopting attention mechanisms to capture global-local dependencies that are more resistant to noise. These findings emphasize that adversarial defense in face swap detection is not limited to explicit adversarial training, but can also be achieved through architectural innovations and targeted feature engineering.

• Explainability

As face swap detection systems grow in complexity and accuracy, the ability to interpret and trust their decisions becomes increasingly important, particularly

in high-stakes environments such as journalism, law enforcement, and social media moderation. Explainable AI (XAI) strategies aim to bridge the gap between black-box model predictions and human-understandable reasoning, enabling users to not only detect manipulated content but also understand why a given frame is flagged as fake.

A notable contribution to this domain is the hierarchical explainable forensics algorithm presented in [101], which integrates human-in-the-loop decision-making with pixel-level visual explanations. This method incorporates an ensemble of detection models, standard and attention-based, augmented by explainability mechanisms such as attention maps, Grad-CAM, and feature attribution techniques. The model first applies deep learning to detect potential manipulations, then presents its decision alongside a set of forensic cues highlighting the specific regions of the face that influenced the outcome.

To generate these cues, the model utilizes attention blocks during training to enforce decision diversity. These blocks encourage the model to make decisions based on multiple facial regions by alternately dropping or enlarging focal regions during training, which both improves generalization and enhances interpretability. During inference, Grad-CAM (Gradient-weighted Class Activation Mapping) is used to backpropagate the model's decision to the last convolutional and attention layers, yielding a heatmap overlaid on the original image. This map visually identifies the areas most responsible for the model's prediction.

The architecture supports pixel-level attribution, where the gradient is used to determine how perturbations to individual pixels affect the prediction outcome. The resulting heatmaps, constructed by weighting and aggregating feature maps and attention responses, highlight key facial regions, such as the eyes or mouth, which are often the most telling indicators of forgery. These maps are further processed via frequency and statistical analysis to support human interpretation.

Such approaches underscore the critical role of explainability in face swap detection. They not only improve trust in automated systems but also facilitate collaboration between machine and human analysts. By providing insight into the internal logic of deep models, explainability enables forensic practitioners to validate detections, understand model failures, and comply with emerging standards for transparent AI systems.

• Lightweight Models for Edge Deployment

Deploying face swap detection models in real-world applications often requires operating under limited computational resources, such as on mobile devices, surveillance systems, or edge computing environments. Therefore, achieving a balance between detection accuracy and model efficiency has become a critical goal in the field. While many state-of-the-art models attain

high performance, their large model sizes and heavy computational demands render them impractical for real-time or on-device deployment.

An exemplary advancement in this direction is the Lightweight Dynamic Fusion Network (LDFNet), which is explicitly designed to offer high detection accuracy with significantly reduced model complexity [53]. LDFNet introduces several innovative, low-complexity components to meet the constraints of edge environments. First, it uses a Spatial Group-wise Enhance (SGE) module to enhance local visual artifacts, key indicators of forgery, while simultaneously suppressing noise. This allows the model to focus on semantically significant features that are crucial for robust detection. To address global context understanding with minimal overhead, the model incorporates a novel Manipulation Trace Extraction Block (TraceBlock), replacing traditional convolution operations. This enables global texture inference, which is essential for detecting subtle manipulations, and is further extended through the Global Texture Representation (GTR) network to hierarchically capture forged regions across different scales. Finally, a Dynamic Fusion Mechanism (DFM) integrates both local and global cues, producing a discriminative and compact feature representation.

LDFNet demonstrates impressive results across several benchmarks (FF++, DFDC, and CelebDF), while requiring only 963K parameters and 801 million floating point operations (FLOPs), where FLOPs refer to the number of arithmetic operations the model performs during inference. Remarkably, LDFNet not only offers superior detection accuracy but also does so with far less computational overhead compared to larger models. In their study, the authors of LDFNet compare its performance against several baseline models, including GoCNet [186], a well-regarded architecture known for its high detection accuracy. GoCNet achieves strong results but uses 28.71 million parameters and 4.76 billion FLOPs, more than 29 times the parameters and nearly 6 times the computational cost of LDFNet. Despite this complexity, LDFNet still outperforms GoCNet, highlighting its excellent trade-off between accuracy and efficiency. This makes LDFNet a strong candidate for practical deployment in resource-constrained environments, where latency, energy efficiency, and storage are major concerns.

Such advancements underscore the importance of model compression, efficient architecture design, and intelligent feature fusion strategies in advancing face swap detection beyond the lab and into real-time, on-device applications.

• Privacy-Preserving Training

As face swap detection systems advance, concerns surrounding user data privacy have grown significantly, especially in scenarios involving sensitive, non-public

datasets. In real-world deployments, data from multiple sources often cannot be centralized due to ethical, legal, or organizational constraints. Traditional training approaches that rely on public, centralized datasets may therefore fail to meet modern privacy expectations. To address this, recent innovations have begun to explore privacy-preserving learning strategies that ensure sensitive user data never leaves local devices. These strategies are not only critical for regulatory compliance but also for fostering public trust in detection tools, particularly in high-stakes applications such as social media monitoring, forensic investigations, and digital identity protection.

One promising direction is the integration of federated learning, where a central model is trained collaboratively across multiple decentralized clients without collecting the raw data itself. Instead, only model updates such as gradients or weights are shared and aggregated on a global server. This approach preserves privacy while allowing the model to learn from diverse and distributed data sources, improving generalization and reducing overfitting to any single dataset.

A representative example of this approach is the FedForgery framework [47], which combines federated learning with a residual-based detection mechanism. FedForgery employs a variational autoencoder to analyze reconstruction residuals that capture subtle differences between real and forged images. These residuals are passed to a classification network, which outputs the final prediction. The federated learning strategy enables multiple local data centers to collaboratively train the model by sharing only parameter updates with the global server, thereby safeguarding sensitive data. After several rounds of weighted aggregation, the global model is refined without accessing any private client data. This method not only protects user privacy but also enhances the discriminative power of the detection model across various forgery types, including previously unseen artifacts.

FedForgery's success demonstrates the viability of privacy-preserving training for face forgery detection in real-world scenarios where data sharing is restricted. It highlights an important direction for future research, balancing the demands of accuracy, generalization, and privacy.

To fully appreciate the significance of these recent innovations, it is essential to understand the growing impact of face swap technologies in practical, real-world settings. The development of adversarially robust, explainable, and privacy-aware models is not just a technical challenge, it is a necessary response to the rising misuse of deepfakes across various sectors. As detection systems evolve, their relevance is increasingly shaped by how well they address the threats and requirements emerging in political, legal, and societal domains.

As the proliferation of face swap technologies continues to accelerate, real-world scenarios involving deepfake misuse have raised urgent concerns about authenticity, manipulation, and accountability. Deepfakes have been used to spread political disinformation, conduct fraud, and create non-consensual explicit content, prompting governments, media institutions, and forensic agencies to seek reliable detection tools. In political contexts, manipulated videos have surfaced during elections in countries like the United States, intensifying debates around digital misinformation and electoral integrity [187]. Law enforcement agencies are also beginning to explore deepfake detection models to assist in digital evidence triage and authenticity verification, although their courtroom admissibility remains under scrutiny.

While no face swap detection model has yet been formally accepted as evidence in court, increasing emphasis is placed on developing tools that can meet forensic standards. Legal and regulatory frameworks are evolving in parallel. The GDPR mandates strict requirements around data handling and privacy, especially for models trained on biometric data. In the United States, several states have passed legislation criminalizing malicious deepfake usage in electoral and adult content contexts, laying the groundwork for legal applications of detection technology [188].

Collaborative initiatives like the Content Authenticity Initiative (CAI) and Project Origin, involving partners such as Adobe, Microsoft, the BBC, and The New York Times, aim to standardize digital provenance and authentication [189]. These efforts, although not court-certified, serve as essential foundations for building trust in detection outputs and pave the way for future forensic admissibility. As such, deepfake detection tools are gaining strategic importance not only in academic and research domains but also across journalism, digital forensics, and legal policy development.

H. BEST PRACTICES FOR BUILDING AND RELEASING FACE SWAP DETECTION TOOLS

To ensure that face swap detection tools are robust, ethically sound, and suitable for real-world deployment, several best practices should be followed. First, models should be trained and validated using diverse and representative datasets to enhance generalization and mitigate bias across different demographic and content domains. As face swap generation techniques continue to evolve rapidly, it is essential to periodically retrain or fine-tune models using newly collected samples and to incorporate adversarial testing strategies that challenge the model with novel or unseen manipulations.

Lightweight model design is another key consideration, especially when targeting deployment in edge devices such as smartphones, surveillance systems, or media verification tools used in the field. Creating compact architectures with fewer parameters and low computational cost (e.g., reduced FLOPs) enables real-time inference without the need for cloud resources, reducing latency and improving privacy by keeping data local. When releasing such models, it is crucial

to benchmark both performance and resource efficiency, so users can assess suitability for their specific hardware constraints.

Transparency and explainability also play vital roles in building trust. Models that provide interpretable outputs or visual explanations (e.g., attention maps or saliency heatmaps) support user understanding and can be especially important in forensic and legal contexts. Releasing documentation such as Model Cards, detailing training data, evaluation protocols, known failure cases, and intended use, helps ensure that models are used appropriately and responsibly.

Additionally, privacy-preserving approaches should be adopted, particularly when training on sensitive or proprietary datasets. Methods like federated learning or synthetic data generation allow collaboration across institutions without exposing raw data, aligning with emerging privacy regulations such as the GDPR.

Finally, responsible release practices include setting usage boundaries through licensing terms, providing disclaimers about limitations, and avoiding overstated claims about reliability. Developers should also monitor for misuse and consider building mechanisms to detect or limit malicious repurposing of the tools. Together, these practices ensure that face swap detection technologies serve public interest while minimizing ethical, legal, and technical risks.

I. FUTURE DIRECTIONS

As face swap detection research matures, several promising avenues merit deeper exploration to overcome current limitations and anticipate emerging challenges.

- **Diverse and Compliant Dataset Generation:**

Future efforts should focus on constructing datasets specifically tailored to face swap manipulation, incorporating a comprehensive variety of techniques to enhance the training and evaluation of detection models. These techniques should include traditional face swapping methods, identity replacement using deep learning frameworks, and fully synthetic face generation through GANs and diffusion models. To ensure resilience against evolving threats, the dataset should also include adversarially crafted examples, manipulated samples intentionally designed to evade detection systems. Incorporating such challenging cases helps evaluate the robustness of models under attack scenarios. A well-rounded dataset must not only capture the visual diversity of real-world content but also reflect the growing sophistication of forgery techniques, ultimately supporting the development of more accurate and generalizable face swap detection systems.

Equally important is the diversity of image and video sources used in dataset construction. Datasets should draw from a wide spectrum of demographics, lighting conditions, resolutions, and real-world scenarios such as social media footage, surveillance-like settings, and professionally shot content to ensure generalizability.

Care should be taken to avoid bias by including balanced representations across age, gender, ethnicity, and geographic backgrounds. Additionally, it is crucial to maintain a balanced dataset with an equal number of fake and real images to prevent model bias and improve detection reliability.

Crucially, these datasets must be constructed with strict adherence to privacy laws and ethical standards, ensuring that data collection, storage, and sharing comply with regulations such as GDPR and other jurisdiction-specific frameworks.

- **Advanced Domain Adaptation:**

Improving generalization remains critical. Future work should focus on sophisticated domain adaptation techniques that allow models to adapt dynamically to unseen data distributions and new manipulation methods without exhaustive retraining. Self-supervised and unsupervised pre-training approaches hold potential to leverage vast amounts of unlabeled data to learn more robust feature representations.

- **Robust Adversarial Defense Mechanisms:** As attackers develop increasingly sophisticated adversarial methods, there is a need to design detection systems resilient to such perturbations. Research on integrating adversarial training, certified robustness guarantees, and real-time anomaly detection can enhance model security.

- **Explainability and Forensic Interpretability:**

For detection tools to be useful in legal and journalistic settings, their decisions must be interpretable. Future research should advance explainability methods tailored to face swap detection, enabling precise localization of manipulations and facilitating human-machine collaboration.

- **Privacy-Preserving Detection Systems:**

Face swap detection often involves sensitive personal data, raising ethical and legal concerns. Privacy-preserving approaches like federated learning, where models are trained across decentralized data sources without centralizing data, and techniques such as differential privacy can be employed to protect user information. Advancing these methods is essential for creating trustworthy systems that comply with privacy regulations and maintain user confidence.

- **Lightweight Architectures for Edge Deployment:**

As detection tools move closer to real-time applications and on-device usage, there is a growing need for lightweight models that can operate efficiently on resource-constrained environments such as smartphones and embedded systems. Future research should explore model compression techniques, knowledge distillation, neural architecture search, and transformer optimization to reduce latency and memory usage while maintaining robust detection performance.

Addressing these directions will push the frontier of face swap detection research and enable deployment of

trustworthy, scalable solutions in an increasingly complex media landscape.

V. CONCLUSION

Face swap detection has emerged as a vital area of research in response to the growing misuse of face manipulation technologies such as deepfakes. This survey has systematically explored the landscape of face swap detection by addressing key aspects including benchmark datasets, performance metrics, comparative evaluation of detection models, and the major challenges that persist in the field. It also provides a detailed taxonomy of detection methods, offering a structured understanding of spatial, temporal, and spatiotemporal approaches. The analysis of cross-dataset generalization performance underscores the critical need for models that can maintain effectiveness across diverse data distributions and domain shifts. While significant progress has been made in developing effective detection methods, challenges such as adversarial robustness, data imbalance, and legal and ethical implications continue to pose hurdles.

Furthermore, this survey highlights recent innovative directions including adversarial defense strategies, explainability techniques, lightweight models designed for edge deployment, and privacy-preserving training frameworks. Best practices for building and releasing face swap detection tools are discussed, emphasizing the importance of accuracy, robustness, ethical compliance, and practical deployment considerations.

Future research must prioritize the creation of more diverse and realistic datasets, development of models resilient to emerging threats, and adherence to privacy and legal standards. By offering a comprehensive overview and critical insights, this survey aims to support and inspire continued advancements in combating face swap-based misinformation and digital deception. Continued collaboration between researchers, industry stakeholders, and policymakers will be essential to address the evolving challenges posed by synthetic media.

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